AI-Powered Synthetic Scenario Generator

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ABSTRACT—

Network intrusion detection systems can under-detect rare attack categories when training data are highly imbalanced. This paper presents the AI-Powered Synthetic Scenario Generator, a prototype that investigates targeted synthetic tabular data augmentation for the minority Worms category in the UNSW-NB15 dataset. A Conditional Tabular Generative Adversarial Network (CTGAN) is used to generate 500 candidate Worms records. Random Forest classifiers trained on real-only data and on real data augmented with synthetic records are compared using the same held-out test split. In the reported experiment, accuracy changes from 0.9234 to 0.9237, Worms recall rises from 0.2500 to 0.3571, and Worms F1-score rises from 0.3889 to 0.4878; Worms precision decreases from 0.8750 to 0.7692. Macro F1 increases from 0.4960 to 0.5043, while weighted F1 remains nearly unchanged. The findings suggest that targeted augmentation may improve minority-class sensitivity, but the precision trade-off and synthetic-record validity require further investigation. A Streamlit dashboard presents samples and comparative metrics.

INDEX TERMS—

Synthetic data generation, CTGAN, UNSW-NB15, network intrusion detection, class imbalance, Worms attack, Random Forest, data augmentation.

I. INTRODUCTION

Network intrusion detection systems (NIDS) analyze network traffic to identify suspicious or malicious activity. Machine-learning classifiers can support this task, but performance is affected by imbalanced labels: common traffic and attack categories may dominate the training data while rare categories are represented by very few records. A classifier can therefore achieve high overall accuracy while missing a meaningful proportion of a rare attack class.

This project investigates whether synthetic tabular records can improve the representation of the Worms attack category in UNSW-NB15. The workflow trains a baseline Random Forest on real training records, generates 500 candidate Worms samples with CTGAN, appends them to the training data, and evaluates a second Random Forest on the same held-out test set. The project also provides an interactive Streamlit dashboard for exploring synthetic samples and model metrics.

The main contributions are: (1) a targeted synthetic-augmentation workflow for the Worms class; (2) a same-test-set comparison of real-only and augmented Random Forest models; and (3) an experimental comparison with class-weighted and SMOTENC baselines. The results are interpreted using minority-class and aggregate metrics rather than accuracy alone.

II. RELATED WORK

The project report's literature review identifies three recurring directions in intrusion detection: machine-learning classification, GAN-based synthetic data generation for class imbalance, and evaluation on benchmark datasets such as UNSW-NB15. Prior work reviewed in the report includes balancing GANs, CTGAN and other GAN variants, knowledge-informed generation, multi-generator architectures, and hybrid deep-learning classifiers [1]–[6].

These approaches motivate targeted generation but also expose practical concerns. Complex generative architectures can increase compute cost; generating more records does not guarantee realistic feature relationships; and improvements in overall accuracy may hide poor minority-class recall. This work focuses on a comparatively direct experiment: generate samples for one rare category, augment the training set, and measure the downstream classifier's change.

III. DATASET AND PROBLEM FORMULATION

The experiment uses UNSW-NB15, a network-traffic dataset containing normal traffic and multiple attack categories [7]. The working split reported in the notebook contains 143,349 records: 114,679 training records and 28,670 test records. The training data include 143 Worms records in the notebook's final split. The in-house report also describes a class-distribution summary with 130 Worms records; this difference likely reflects different preprocessing or split stages, so the experimental split counts are used here where available.

Let D_train denote the original training set and D_test the held-out test set. The baseline classifier is trained on D_train. CTGAN is conditioned on the Worms category and produces 500 candidate synthetic records, S_Worms. The augmented training set is D_aug = D_train ∪ S_Worms. The baseline and augmented classifiers are evaluated on the same D_test to reduce evaluation differences caused by test-set selection.

The objective is not simply to maximize accuracy; it is to improve the ability to detect Worms attacks while keeping aggregate classification performance stable.

IV. PROPOSED SYSTEM AND METHODOLOGY

A. Preprocessing and split

The notebook loads the UNSW-NB15 CSV files, applies feature names, selects the working feature set, and creates a train/test split. Categorical and numerical attributes are handled through preprocessing pipelines for classification. The final documented experiment uses 114,679 real training records and 28,670 test records. The test set is held constant for baseline and augmented evaluation.

B. CTGAN-based generation

CTGAN is trained using real training examples from the Worms class. The notebook records 143 Worms training records and generates 500 candidate synthetic rows with 41 feature columns. CTGAN is suitable for mixed tabular data because it models categorical and continuous columns. The synthetic records are intended to supplement, not replace, the original training examples.

C. Classifier training

Two Random Forest pipelines are compared: (i) a baseline trained on real training data only and (ii) an augmented model trained on the real training data plus 500 synthetic records. Random Forest is used as the downstream classifier because it provides a practical baseline for tabular classification. The notebook also evaluates class-weighted Random Forest and SMOTENC-based Random Forest as comparison strategies.

D. Dashboard

The Streamlit application presents an overview of the experiment, synthetic sample previews, real-only versus augmented comparisons, evaluation reports and confusion matrices, and downloadable experiment outputs. The deployed interface is available at https://synthetic-network-attack-scenario-generator-7d6fjszp4hpjfgdq72.streamlit.app/.

E. Evaluation metrics

Accuracy measures the fraction of correct predictions. Precision measures the fraction of predicted Worms instances that are correct; recall measures the fraction of actual Worms instances detected; and F1-score is the harmonic mean of precision and recall. Macro F1 weights classes equally, whereas weighted F1 accounts for class support. Confusion matrices support class-wise inspection.

F. Comparison methods and metrics

The experiment compares four approaches: (i) baseline Random Forest, (ii) class-weighted Random Forest, (iii) SMOTENC with Random Forest, and (iv) Random Forest trained on real data plus 500 CTGAN generated Worms records. Accuracy measures the proportion of correctly classified test records. For the Worms class, precision measures the fraction of predicted Worms records that are truly Worms; recall measures the fraction of actual Worms records found; and F1-score is the harmonic mean of precision and recall. Macro F1 gives equal weight to each class, whereas weighted F1 weights each class by its support.

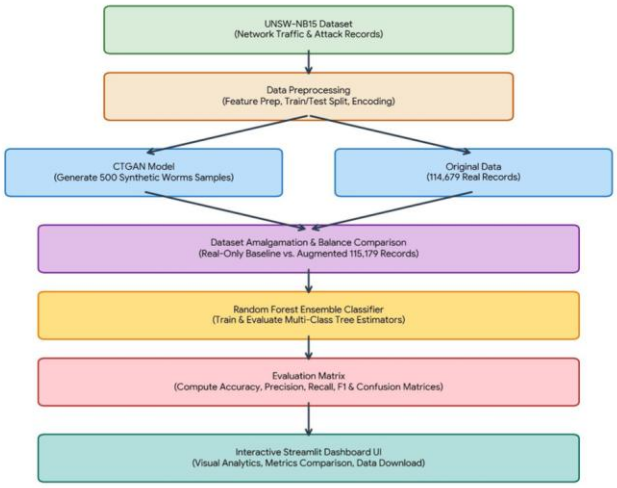


Figure 1. Project pipeline from UNSW-NB15 preprocessing to CTGAN augmentation, Random Forest evaluation and the Streamlit dashboard.

V. EXPERIMENTAL RESULTS

Table I summarizes the reported comparison. Accuracy improves by only 0.0003, while Worms recall increases by 0.1071 and Worms F1-score increases by 0.0989. These correspond to relative increases of approximately 42.8% in recall and 25.4% in F1-score. Worms precision falls by 0.1058, indicating a trade-off: the augmented classifier identifies more Worms instances but also produces a greater proportion of false-positive Worms predictions.

Metric	Real-only RF	Real + synthetic RF
Accuracy	0.9234	0.9237
Worms precision	0.8750	0.7692
Worms recall	0.2500	0.3571
Worms F1-score	0.3889	0.4878
Weighted F1	0.9219	0.9220
Macro F1	0.4960	0.5043

TABLE I. REAL-ONLY AND CTGAN-AUGMENTED RANDOM FOREST RESULTS

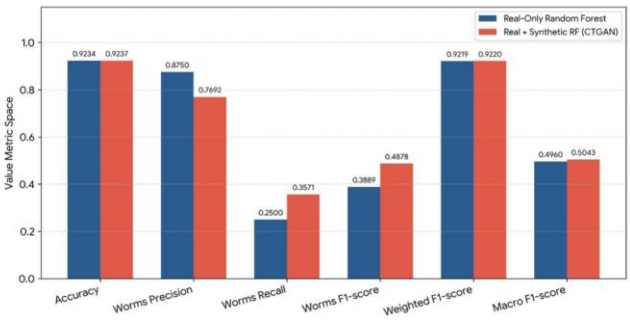


Figure 2. Reported metric comparison between the real-only and CTGAN-augmented classifiers.

The small increase in aggregate accuracy should not be interpreted as a large overall improvement. The more meaningful change for this project is the gain in Worms recall and F1. Macro F1 also rises by 0.0083, suggesting a modest improvement in balanced class-level performance, while weighted F1 remains almost constant.

A. Comparison with alternative balancing methods

Model	Accuracy	Macro F1	Worms recall	Worms F1
Baseline RF	0.9234	0.4960	0.2500	0.3889
Class-weighted RF	0.9223	0.5209	0.5000	0.6364
SMOTENC RF	0.9232	0.5113	0.5357	0.5660
Real + synthetic RF	0.9237	0.5043	0.3571	0.4878

TABLE II. REPORTED BALANCING-STRATEGY COMPARISON

In the reported ablation results, class-weighted Random Forest and SMOTENC achieve higher Worms recall than the CTGAN-augmented model. Class-weighted Random Forest also achieves the highest Worms F1 among these four configurations. Therefore, CTGAN augmentation is a useful experimental option, but the current results do not establish it as the best balancing method across all metrics.

VI. IMPLEMENTATION AND REPRODUCIBILITY

The project uses Python with Pandas and NumPy for data processing, CTGAN for synthetic tabular generation, scikit-learn Random Forest pipelines for classification and metrics, imbalanced-learn for SMOTENC comparisons, Matplotlib for visualizations, and Streamlit for the dashboard. The provided notebook documents the data inspection, split, generation, model training and comparison workflow. The README describes the dashboard sections and downloadable experiment outputs.

The same held-out test set is used for the principal baseline-versus-augmented comparison. However, the reported results represent one experimental setup and

split; the README explicitly notes that metrics can vary with preprocessing, generated samples, model settings and random seeds.

VII. Streamlit Prototype

The project exposes the experiment through a Streamlit dashboard. According to the supplied README, the interface contains a main dashboard, synthetic-sample preview, real-versus-augmented comparison, model-evaluation views with classification reports and confusion matrices, and a section for downloading results. The live prototype is available at:

https://synthetic-network-attack-scenario-generator 7d6fjszp4hpjfgdq72.streamlit.app/

The dashboard provides an accessible way to inspect outputs; it does not by itself establish model validity or production readiness. The experimental conclusions in this paper are based on the supplied notebook and report, not on an independent audit of the live deployment.

VII. LIMITATIONS AND DISCUSSION

A key limitation concerns synthetic-data validity. The notebook's validation cells report negative values in multiple generated features and, in one filtering pass, all 500 generated rows were removed as invalid. A later revised generation pass still reports out-of-range or negative values in some features. This means the reported classification improvement should be interpreted cautiously: the paper's metrics document an augmentation experiment, but additional feature constraints and validation are needed to demonstrate that all generated records are realistic and valid.

Other limitations include evaluation on a single split, the small number of real Worms examples, potential false positives reflected in reduced precision, and the absence of repeated-seed confidence intervals. The current results do not prove generalization to other datasets, unseen attack families, or live network traffic. The test set should remain untouched during generator tuning and all synthetic generation should be confined to training data to avoid leakage.

A further consistency issue in the report is that one table lists 82,332 test records, whereas the notebook and experimental narrative report 28,670. This paper uses 28,670 because it is supported by the notebook's explicit split output and the 114,679 + 28,670 = 143,349 working dataset total.

VIII. CONCLUSION AND FUTURE WORK

This paper presented the AI-Powered Synthetic Scenario Generator, a prototype for investigating targeted synthetic data augmentation in network intrusion detection. Using UNSW-NB15, the experiment generated 500 candidate Worms records with CTGAN and compared Random Forest models trained on real-only and augmented data. The augmented model increased Worms recall from 0.2500 to 0.3571 and Worms F1-score from 0.3889 to 0.4878, while accuracy remained nearly unchanged and Worms precision decreased. Compared with class weighting and SMOTENC, CTGAN did not achieve the strongest minority-class metrics in the reported experiment.

Future work should enforce domain-aware feature ranges and integer constraints, assess statistical similarity and downstream utility of synthetic rows, compare augmentation ratios and CTGAN variants, repeat experiments across stratified splits and random seeds, and evaluate additional classifiers and attack categories. Integration with live monitoring should be considered only after synthetic-data validity, robustness and leakage controls are established.

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